Al Imam Mohammad Ibn Saud Islamic University
College of Computer and Information Sciences

Computer Science Department

CS1213-Design and Analysis of Algorithms		1st Semester 2025/2026		Quiz# 1
Student Name	[redacted]	ID:	[redacted]	
CLOs:	1.3	Grade:	[4] / 10	

(PLO#K1-CLO1.3) Question 1: To be answered in **(20)** Minutes [ ] / 10 Marks

Choose one correct answer for each statement and fill the following table: [1 point for each item]

Question	1	2	3	4	5	6	7	8
Answer	A	b	c	b	d	A	c	A

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1. if $f(n) = 5.2\,n^3 + 20n + 1$ and $g(n) = 1.5\,n^3$, then:
   a. $f(n) = O(g(n))$ —
   b. $f(n) = \Omega(g(n))$
   c. $f(n) = \Theta(g(n))$
   d. All the above answers are correct.

2. What is the largest input size n that we can get it within a day if we have an algorithm with the running time n and a computer with the performance 1000 operations per hour?
   a. $n = 1000$
   b. $n = 24000$ —
   c. $n = 3600$
   d. $n = 86400$

3. Given the functions $f(n) = 7n^5 + n$ and $g(n) = 5n^3$ which values of c and n_0 can be used to prove that $f(n) = \Omega(g(n))$?
   $7n^5 + n \geq 5n^3$
   a. $c = 5, n_0 = 2$
   b. $c = 100, n_0 = 2$
   — c. $c = 10, n_0 = 1$
   d. All the above answers are correct.

4. If $f(n) = n^3 + n$, then $f(n)$ is:
   a. $O(n^3 + n)$
   — b. $o(n^3 + n)$
   c. $\omega(n^3 + n)$
   d. All the above answers are incorrect.

5. If f(n) is not O(g(n)) then:
   a. $f(n) = O(g(n))$
   b. $f(n) = \Omega(g(n))$
   c. $f(n) = \Theta(g(n))$
   d. Cannot be determined.

6. Consider two different algorithms A and B. Both algorithms have been implemented using the same programming language. Using inputs with the same size, if the running time of A is 0.5 millisecond while the running time of B is 1 millisecond, then:
   a. A is faster than B
   b. B is faster than A
   c. A and B have the same speed
   d. Cannot be determined

7. If $f(n) = (2n^7 + n^4 + 10n^3)(3m^4 + 2m^3 + 1)$, then:
   a. $f(n) = O(n^{11})$
   b. $f(n) = O(n^7 m^4)$
   c. $f(n) = O(n^7)$
   d. All the above answers are incorrect.

8. If $f(n) = 3^n$ and $g(n) = \sqrt{n}^n$, then
   a. f grows faster than g.
   b. f grows slower than g.
   c. f and g have the same rate of growth.
   d. Cannot be determined.

(PLO#K1-CLO1.3) Question 2: To be answered in **(10) Minutes** [ ] / 2 Marks

What is the time complexity of the following algorithm segments in the most precise Big O? [1 point for each item]

Segment A	Segment B

`j ← 1`
`for j<=n do`
`  k ←1`
`  for k<=j do`
`  print j`
`  k ←k+1`
`  endFor`
`  j ←j+1`
`endFor`
	`i ←1`
	`while i< n do`
	`  if 0 < n`
	`  j ←1`
	`  for j<= n² do`
	`  j ←i+1`
	`  endFor`
	`  else`
	`  print n`
	`  endIf`
	`  i ←i*2`
	`endWhile`

Answer: $O(n^2)$	Answer: $O(n^2 \log n)$